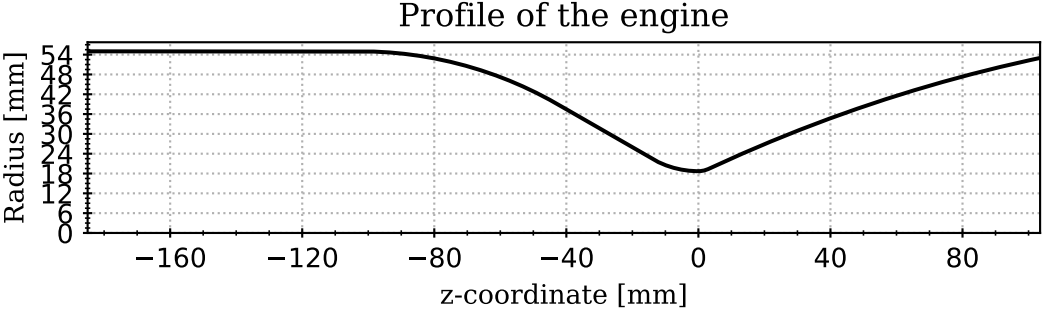
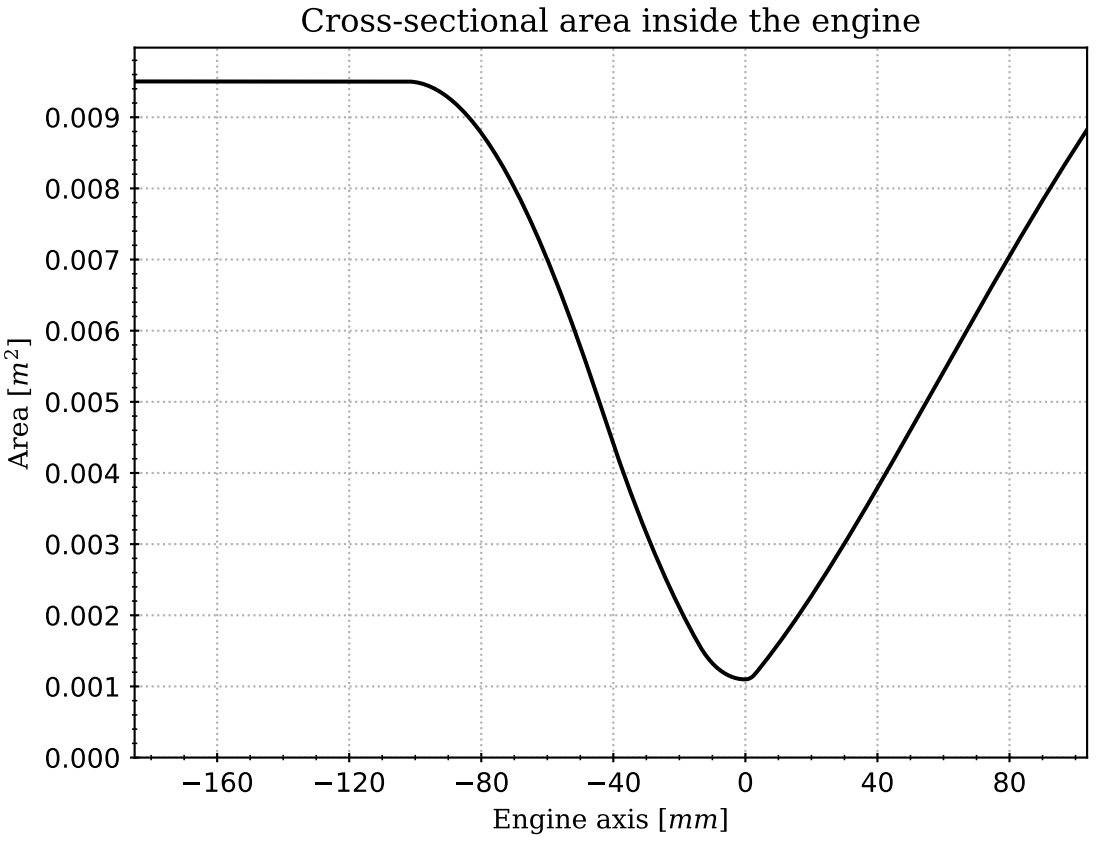
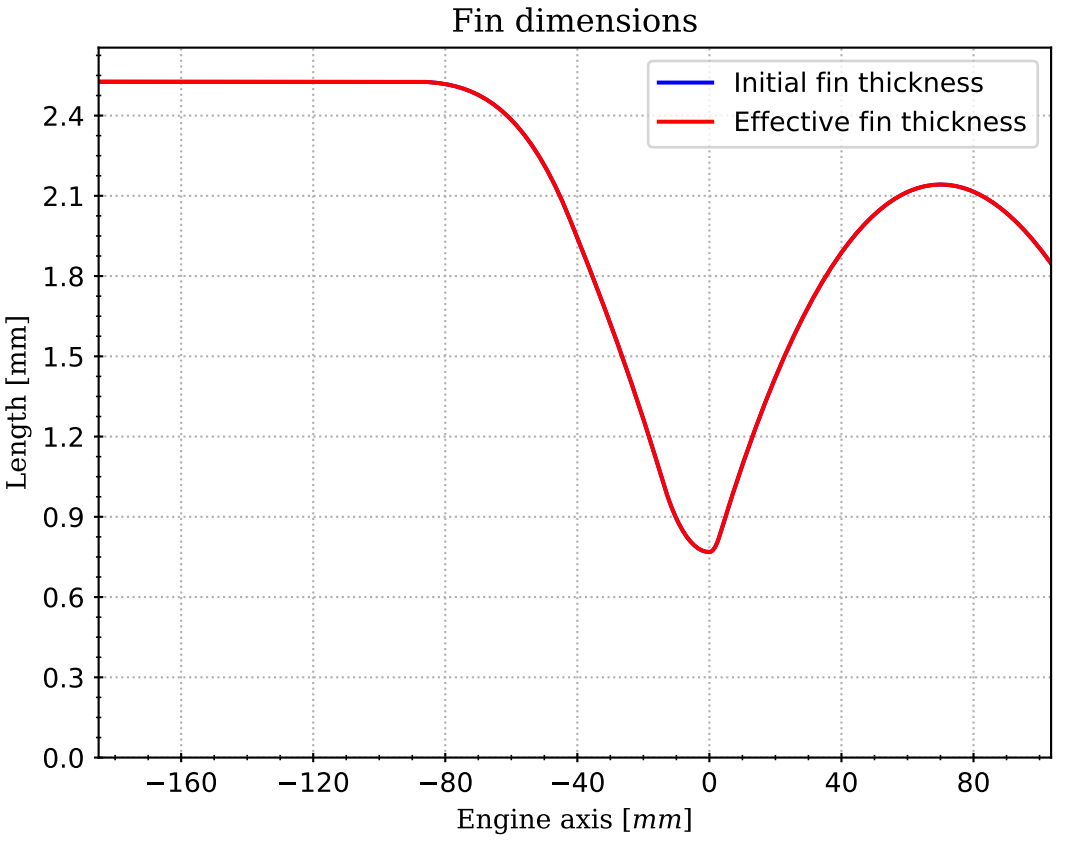
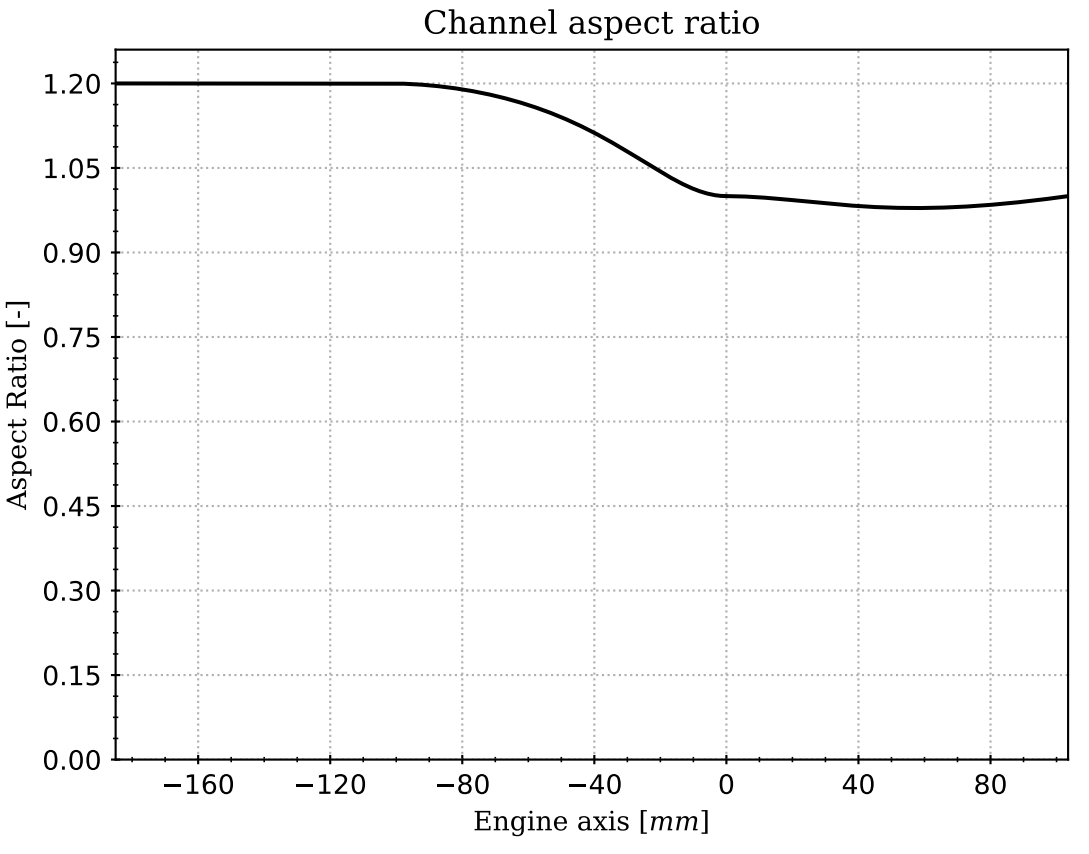
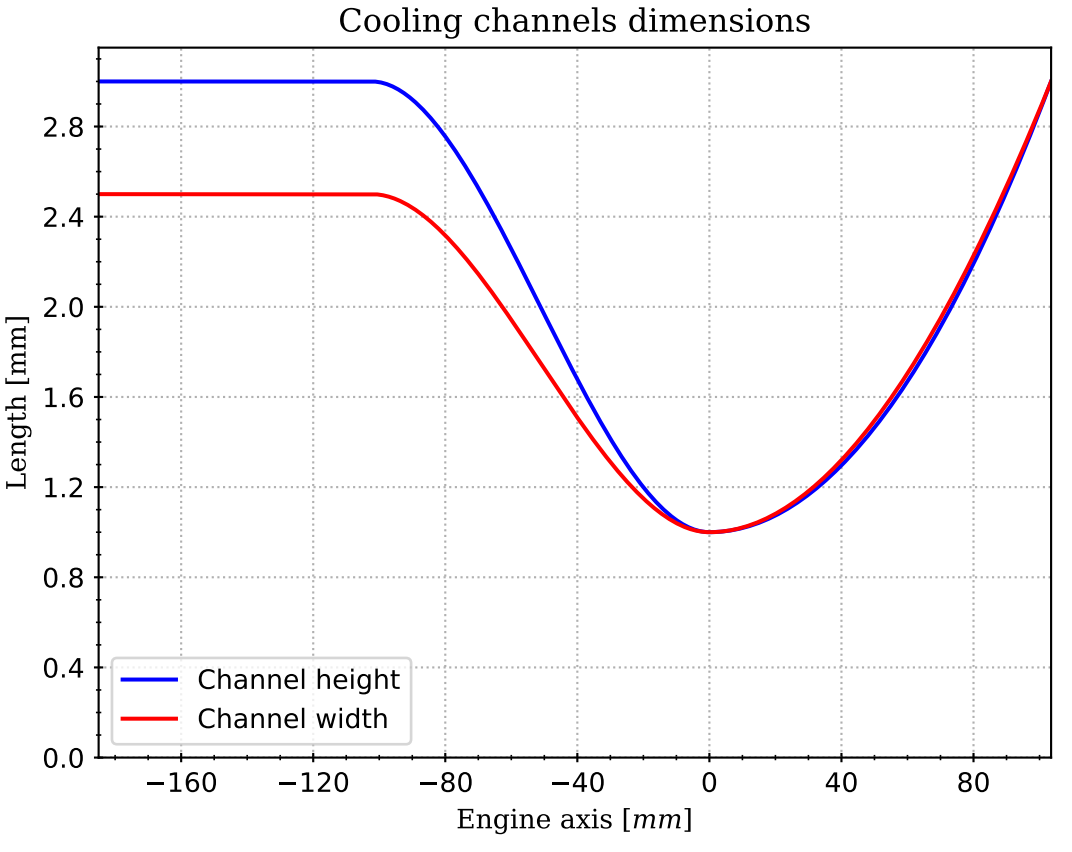
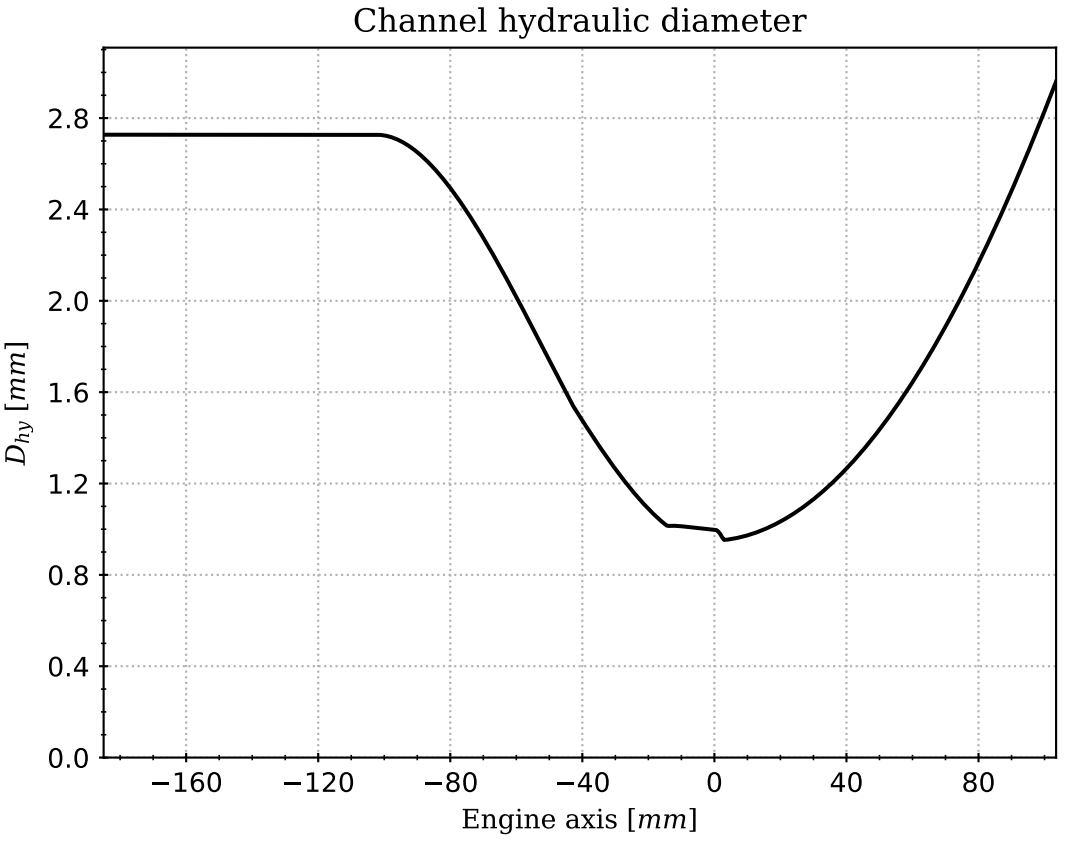

Profile of the engine
Radius [mm]
54
48
42
36
30
24
18
12
6
0
−160
−120
−80
−40
0
40
80
z-coordinate [mm]

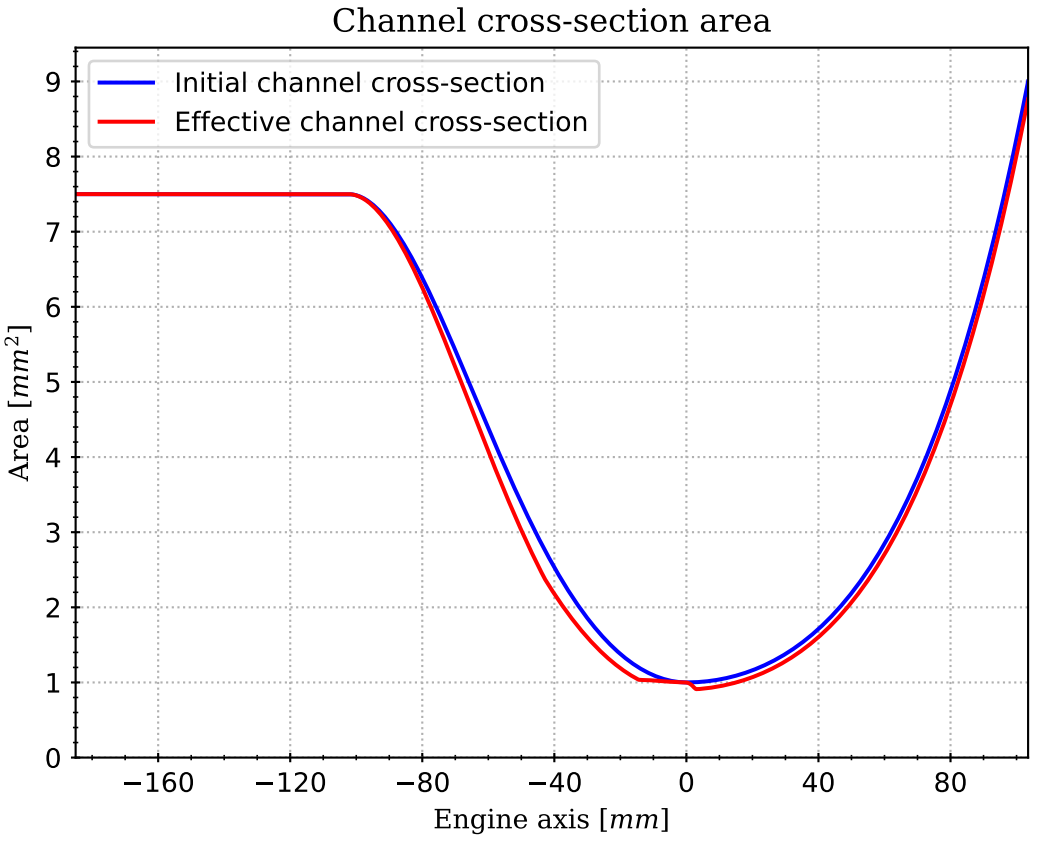

Cross-sectional area inside the engine
Area [m^2]
0.000
0.001
0.002
0.003
0.004
0.005
0.006
0.007
0.008
0.009
−160
−120
−80
−40
0
40
80
Engine axis [mm]

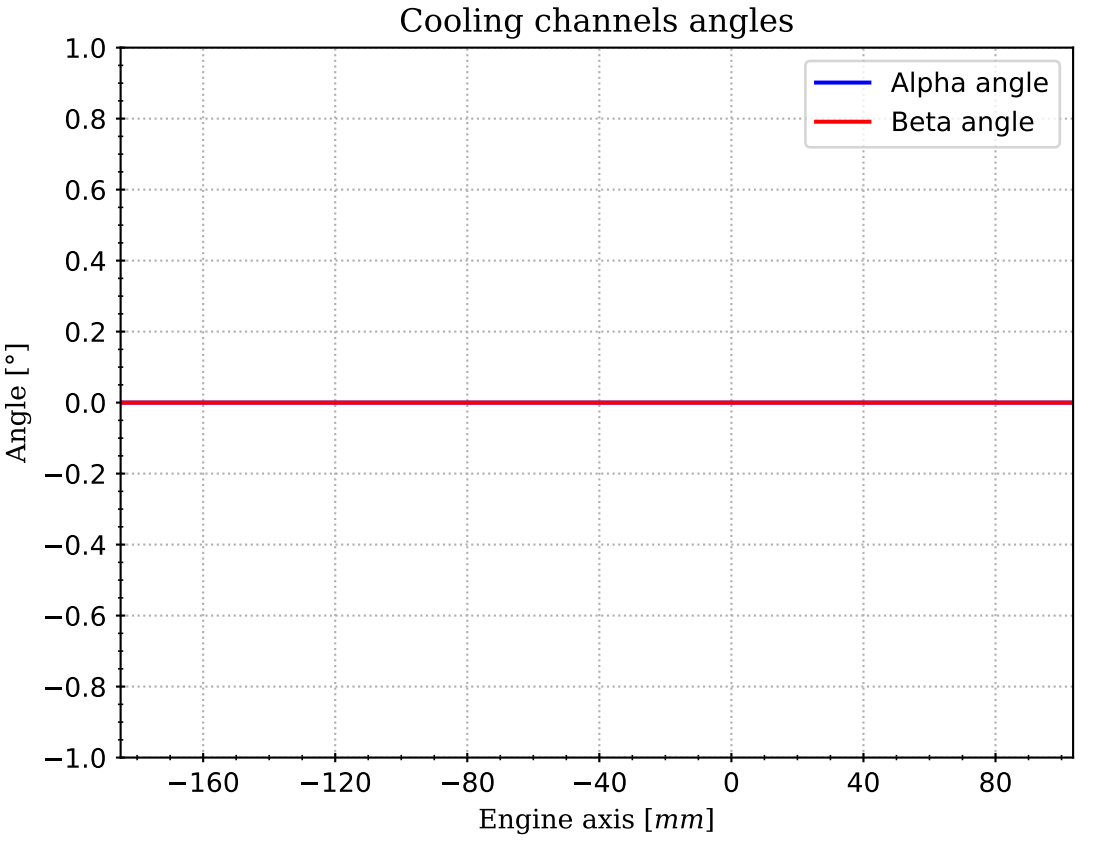

Fin dimensions
Initial fin thickness
Effective fin thickness
Length [mm]
Engine axis [mm]
0.0
0.3
0.6
0.9
1.2
1.5
1.8
2.1
2.4
−160
−120
−80
−40
0
40
80

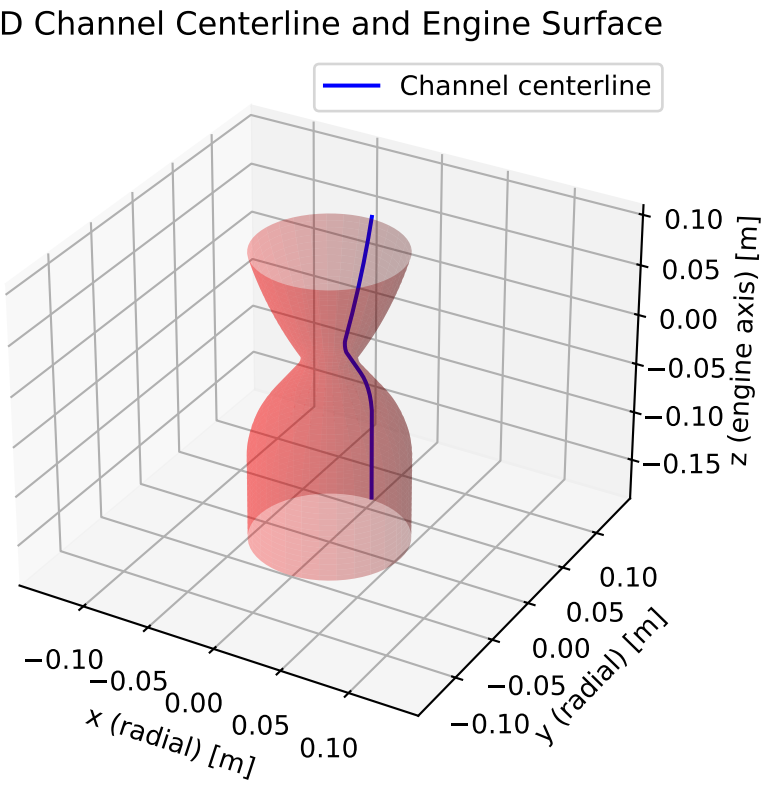

Channel aspect ratio
1.20
1.05
0.90
0.75
0.60
0.45
0.30
0.15
0.00
Aspect Ratio [-]
−160
−120
−80
−40
0
40
80
Engine axis [mm]

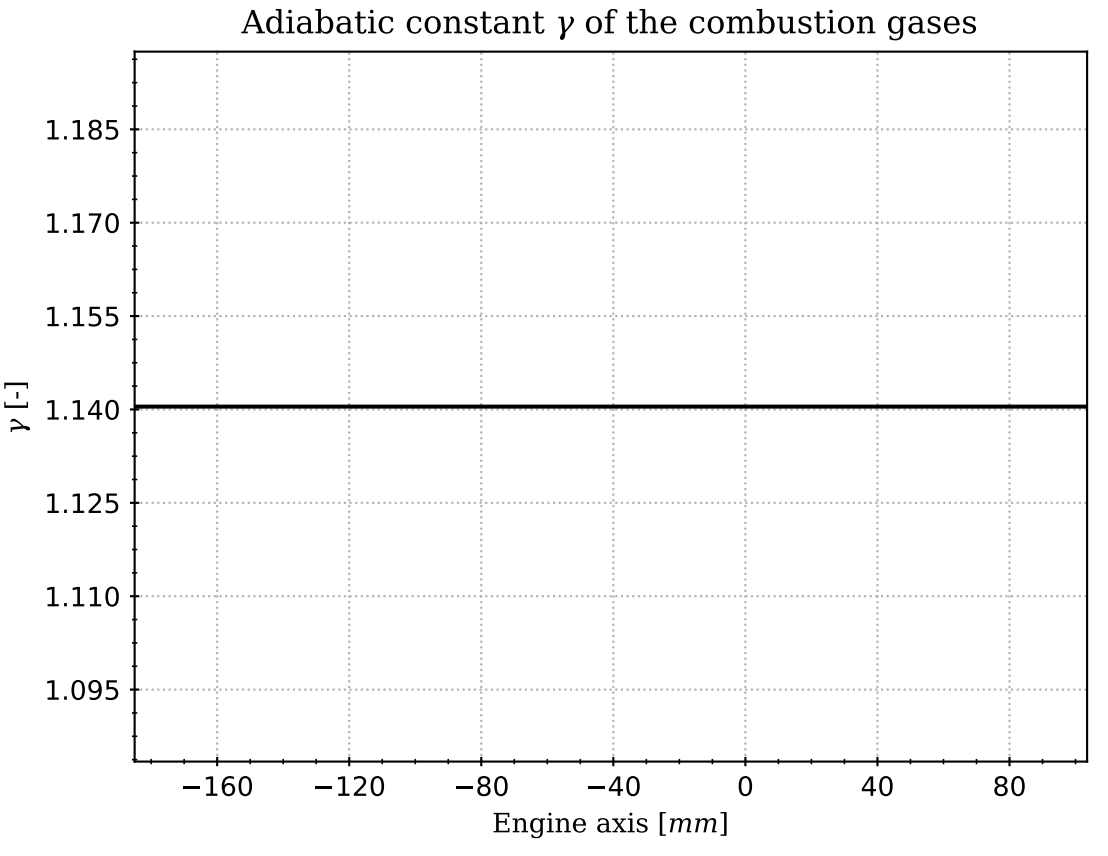

Cooling channels dimensions
Length [mm]
Engine axis [mm]
Channel height
Channel width
0.0
0.4
0.8
1.2
1.6
2.0
2.4
2.8
−160
−120
−80
−40
0
40
80

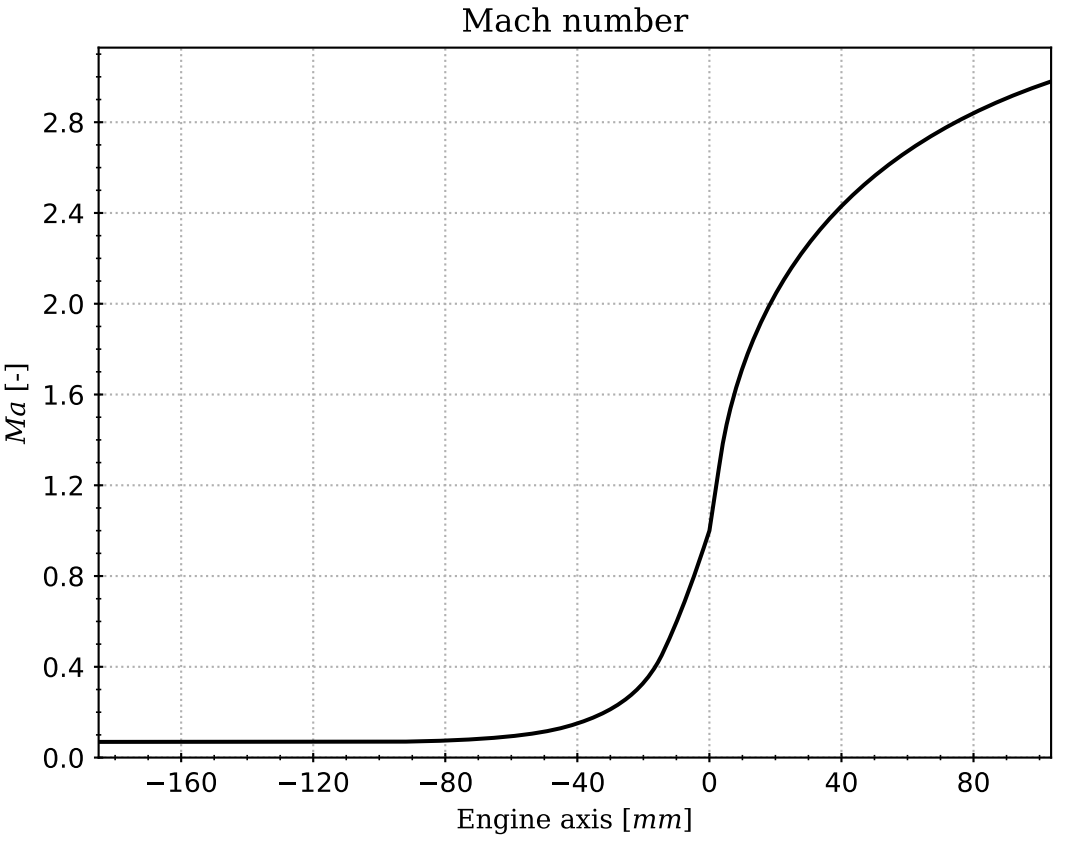

Channel hydraulic diameter
D_{hy} [mm]
2.8
2.4
2.0
1.6
1.2
0.8
0.4
0.0
−160
−120
−80
−40
0
40
80
Engine axis [mm]

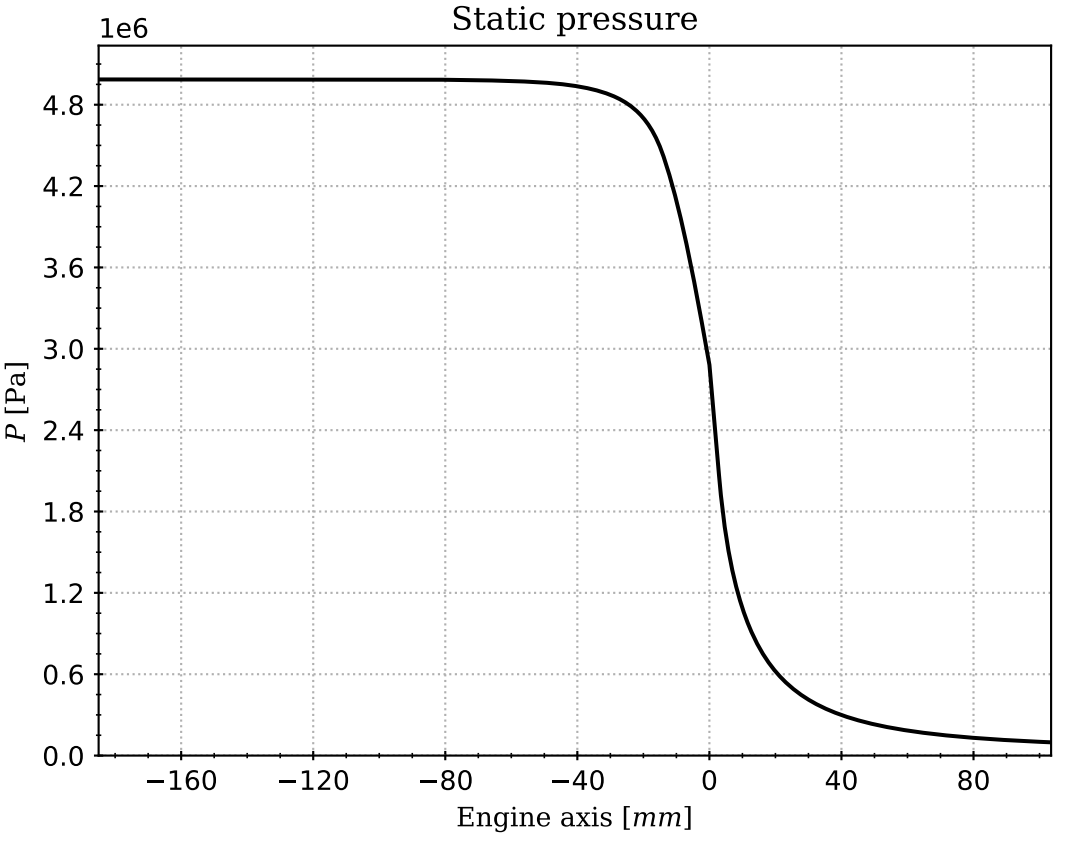

Channel cross-section area
Initial channel cross-section
Effective channel cross-section
Area [mm^2]
Engine axis [mm]
0
1
2
3
4
5
6
7
8
9
−160
−120
−80
−40
0
40
80

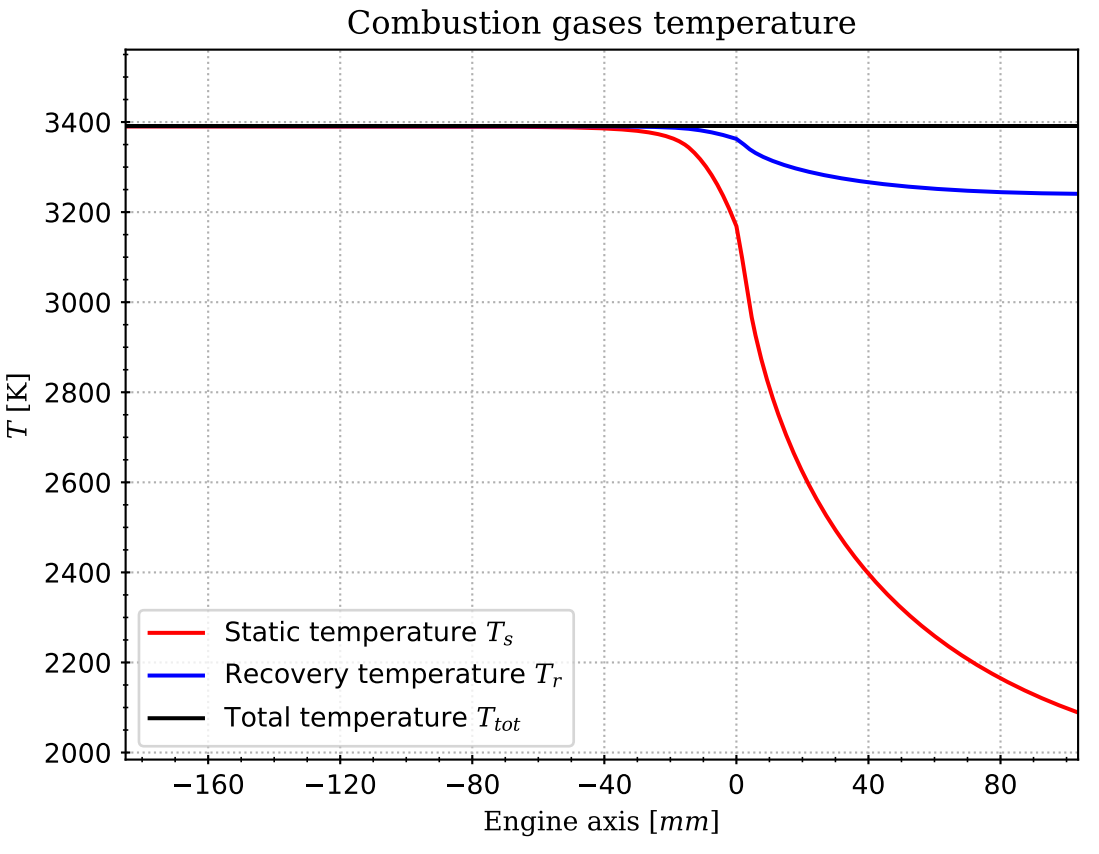

Cooling channels angles
Alpha angle
Beta angle
Angle [°]
Engine axis [mm]
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−160
−120
−80
−40
0
40
80

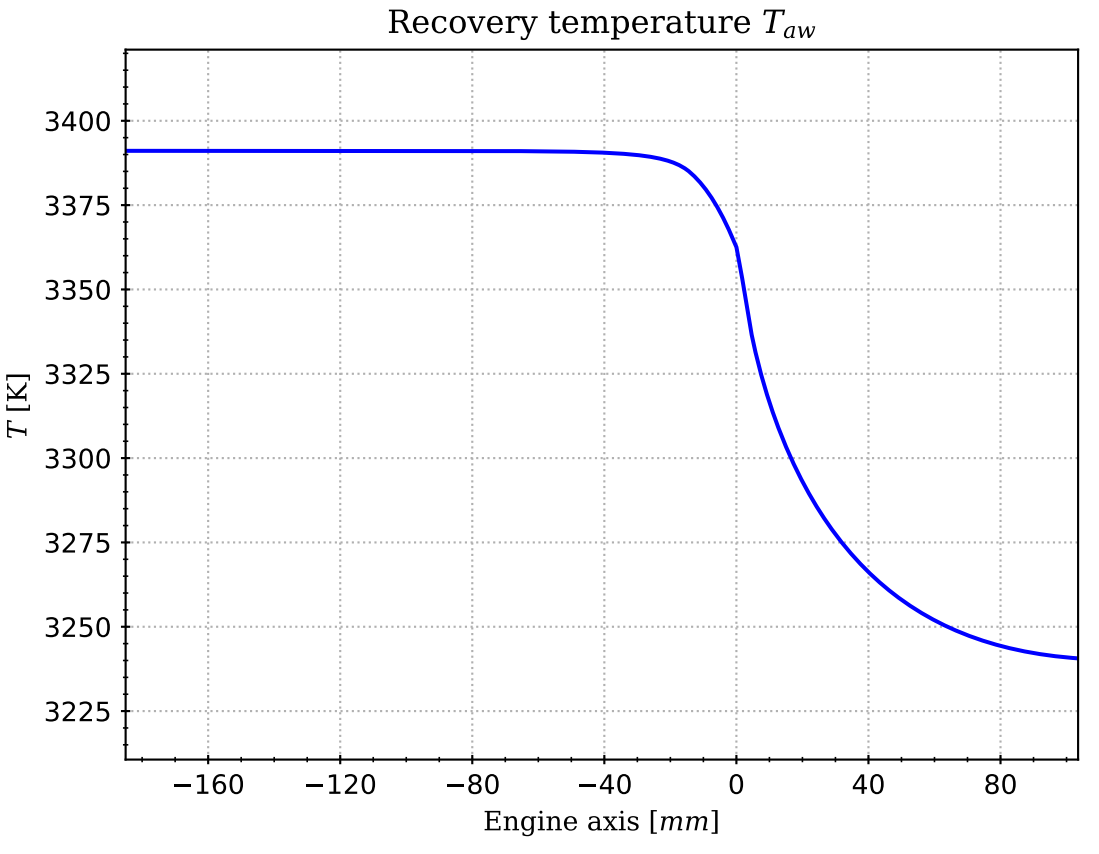

3D Channel Centerline and Engine Surface
Channel centerline
0.10
0.05
0.00
−0.05
−0.10
−0.15
z (engine axis) [m]
0.10
0.05
0.00
−0.05
−0.10
y (radial) [m]
−0.10
−0.05
0.00
0.05
0.10
x (radial) [m]

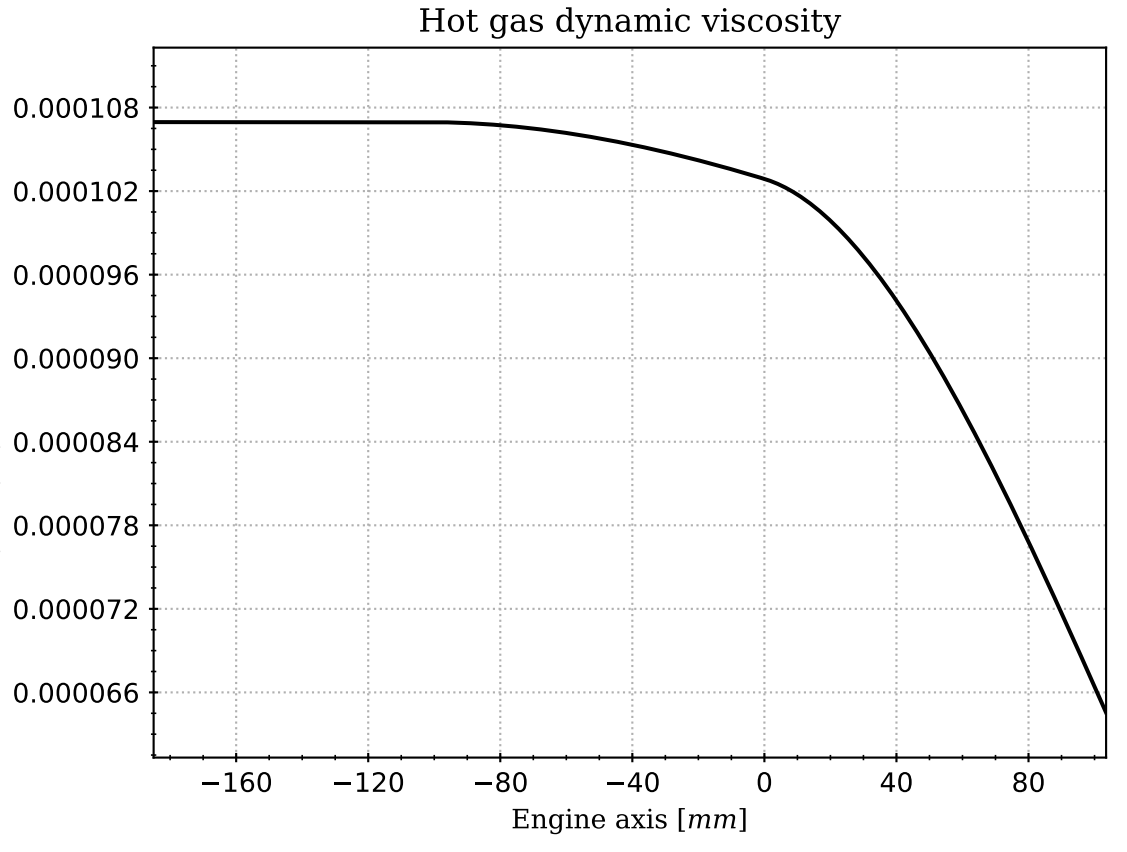

Adiabatic constant γ of the combustion gases
1.185
1.170
1.155
1.140
1.125
1.110
1.095
γ [-]
−160
−120
−80
−40
0
40
80
Engine axis [mm]

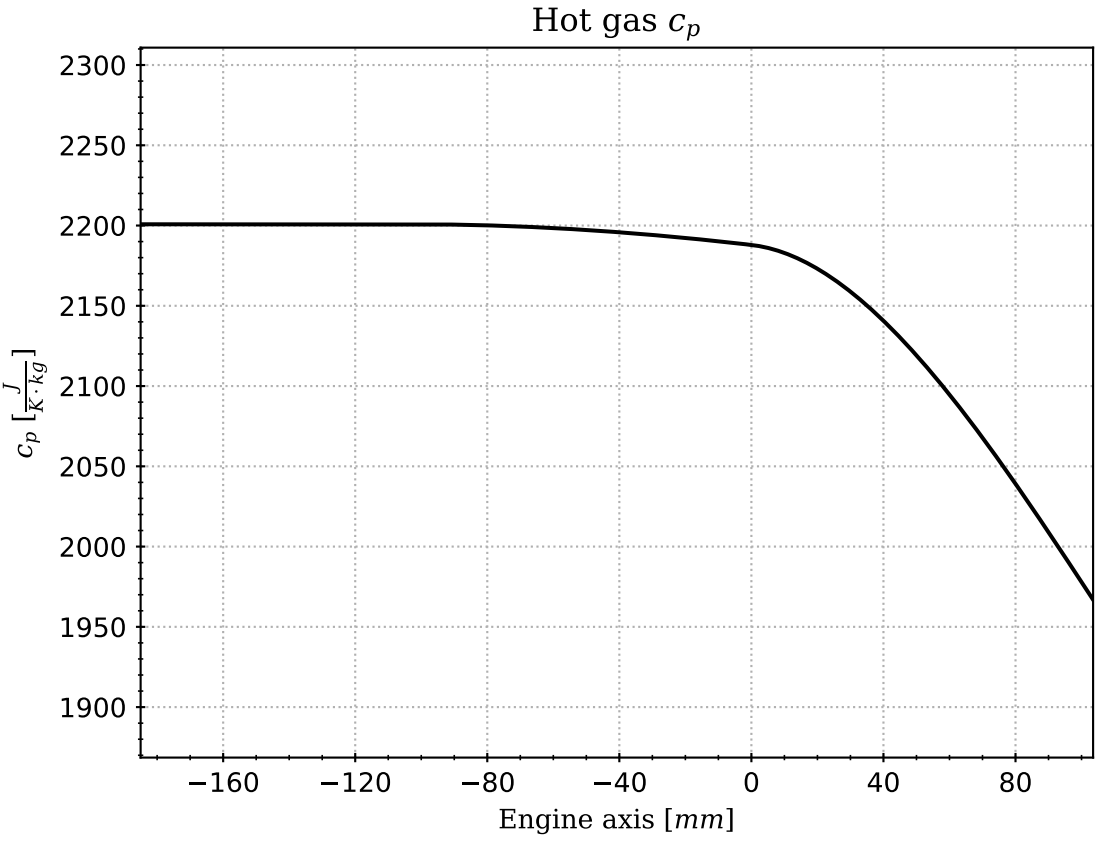

Mach number
2.8
2.4
2.0
1.6
1.2
0.8
0.4
0.0
Ma [-]
−160
−120
−80
−40
0
40
80
Engine axis [mm]

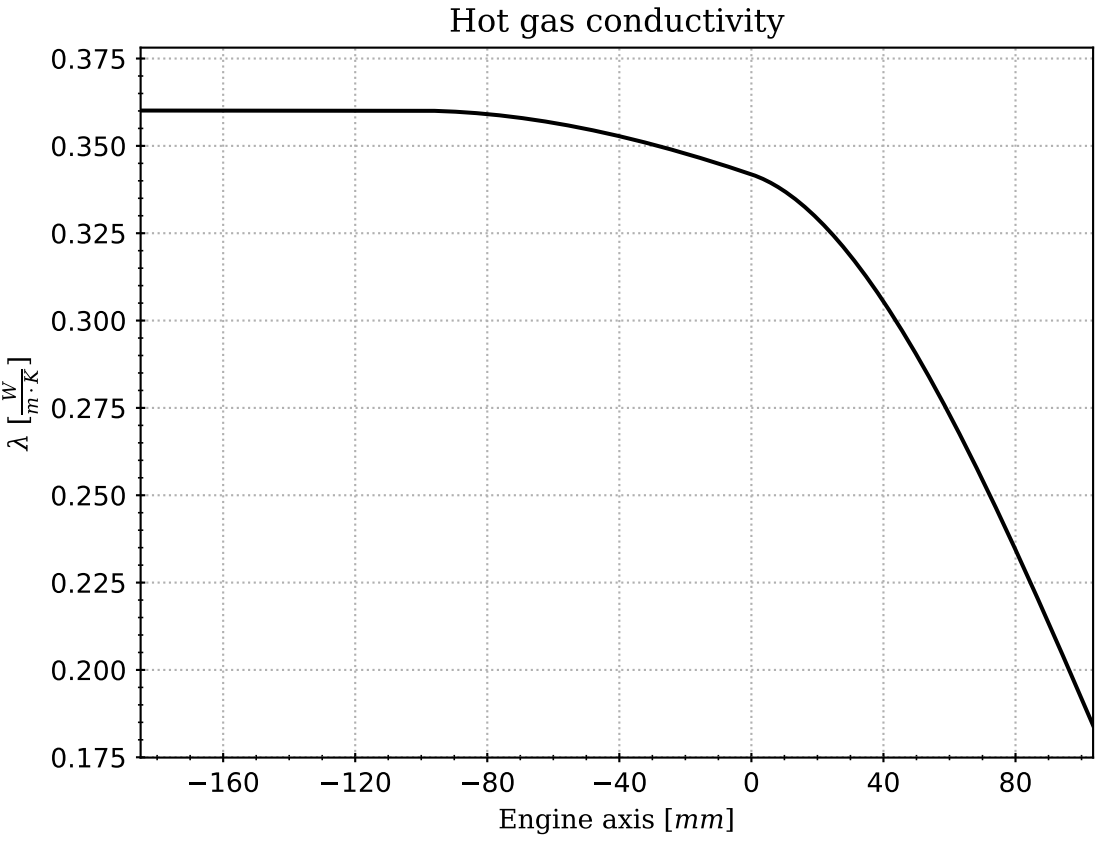

Static pressure
1e6
4.8
4.2
3.6
3.0
2.4
1.8
1.2
0.6
0.0
P [Pa]
−160
−120
−80
−40
0
40
80
Engine axis [mm]

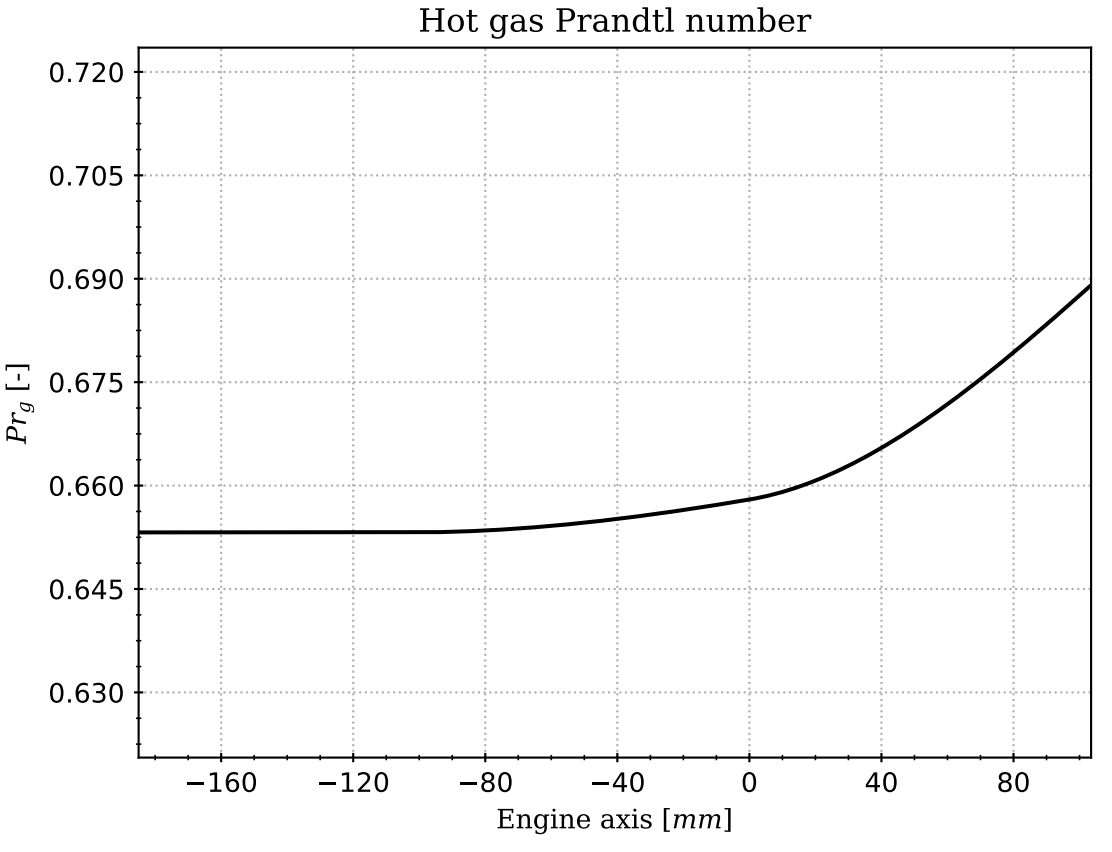

Combustion gases temperature
3400
3200
3000
2800
2600
2400
2200
2000
T [K]
−160
−120
−80
−40
0
40
80
Engine axis [mm]
Static temperature T_s
Recovery temperature T_r
Total temperature T_{tot}

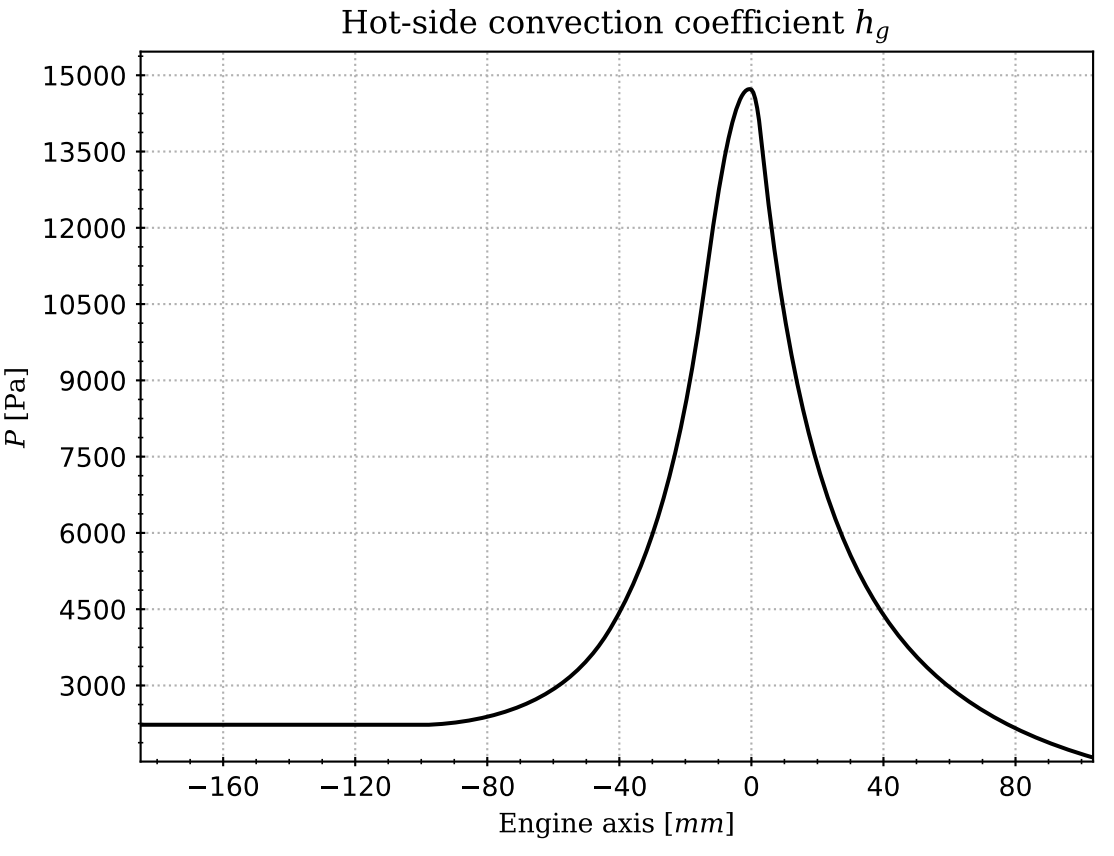

Recovery temperature T_{aw}
3400
3375
3350
3325
3300
3275
3250
3225
T [K]
−160
−120
−80
−40
0
40
80
Engine axis [mm]

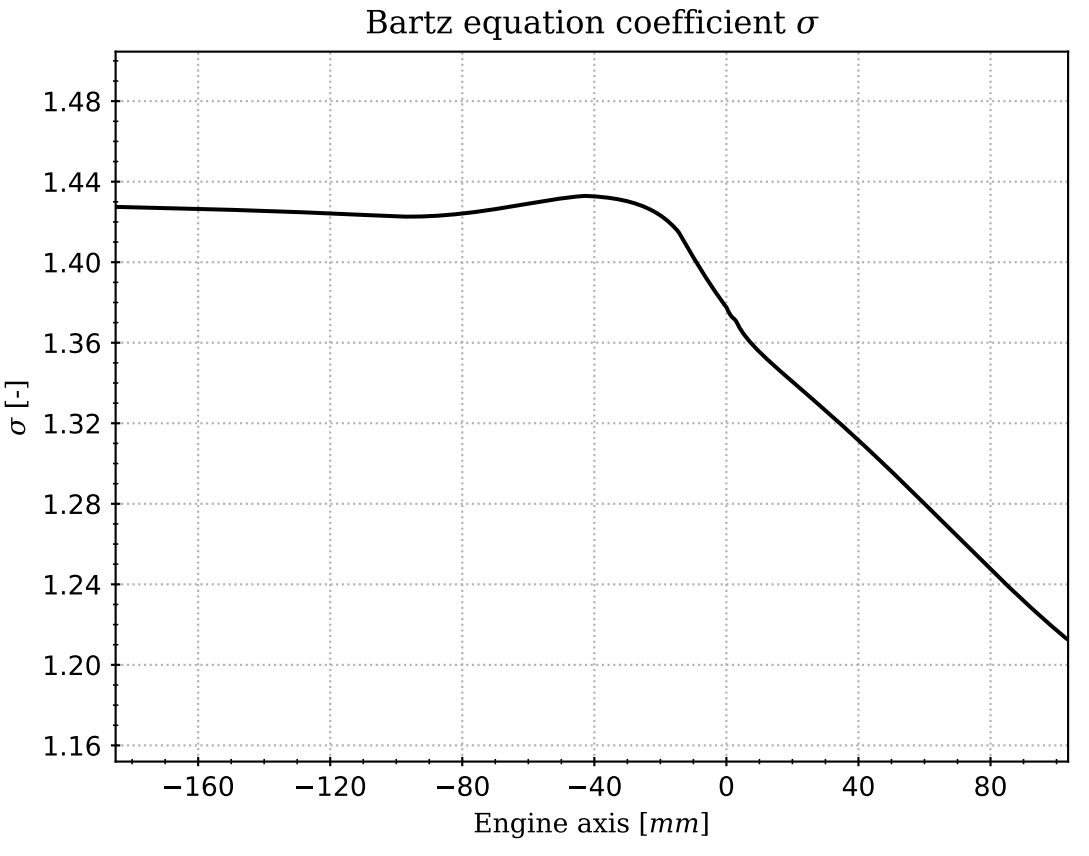

Hot gas dynamic viscosity
0.000108
0.000102
0.000096
0.000090
0.000084
0.000078
0.000072
0.000066
−160
−120
−80
−40
0
40
80
Engine axis [mm]

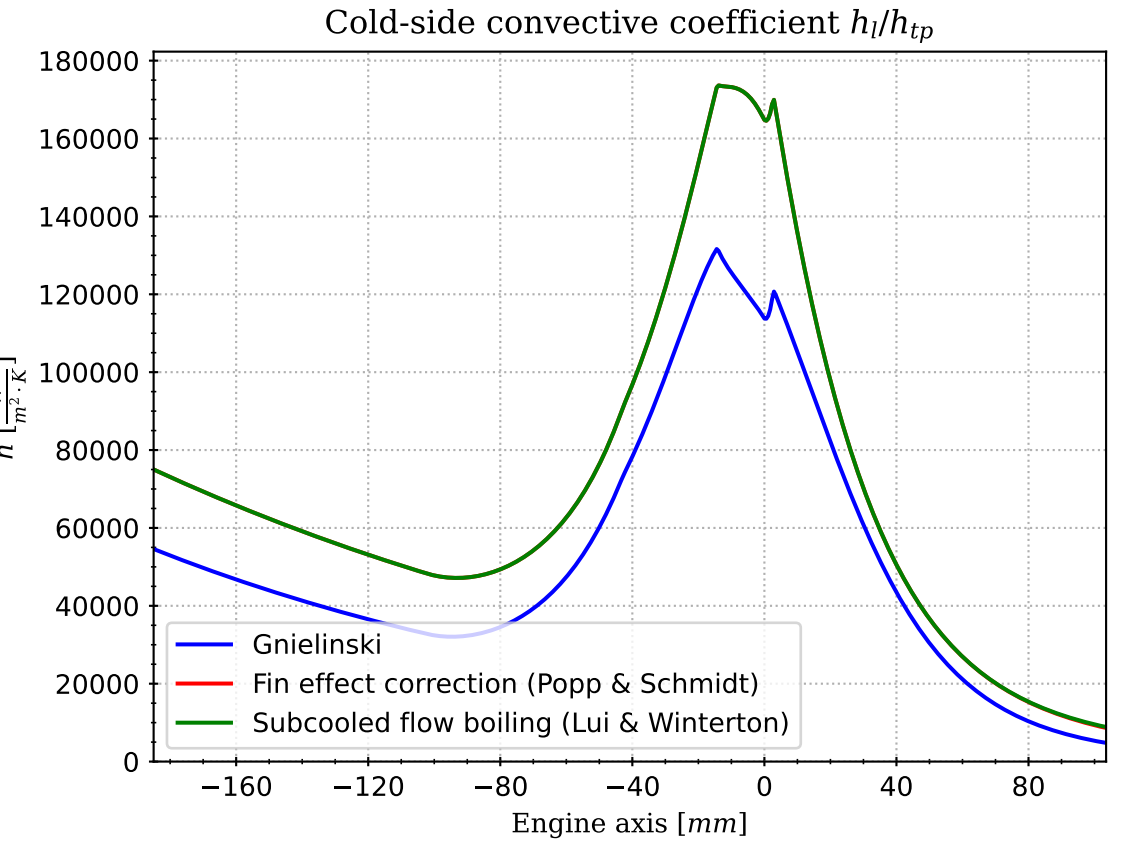

Hot gas c_p
2300
2250
2200
2150
2100
2050
2000
1950
1900
c_p $[\frac{J}{K \cdot kg}]$
−160
−120
−80
−40
0
40
80
Engine axis $[mm]$

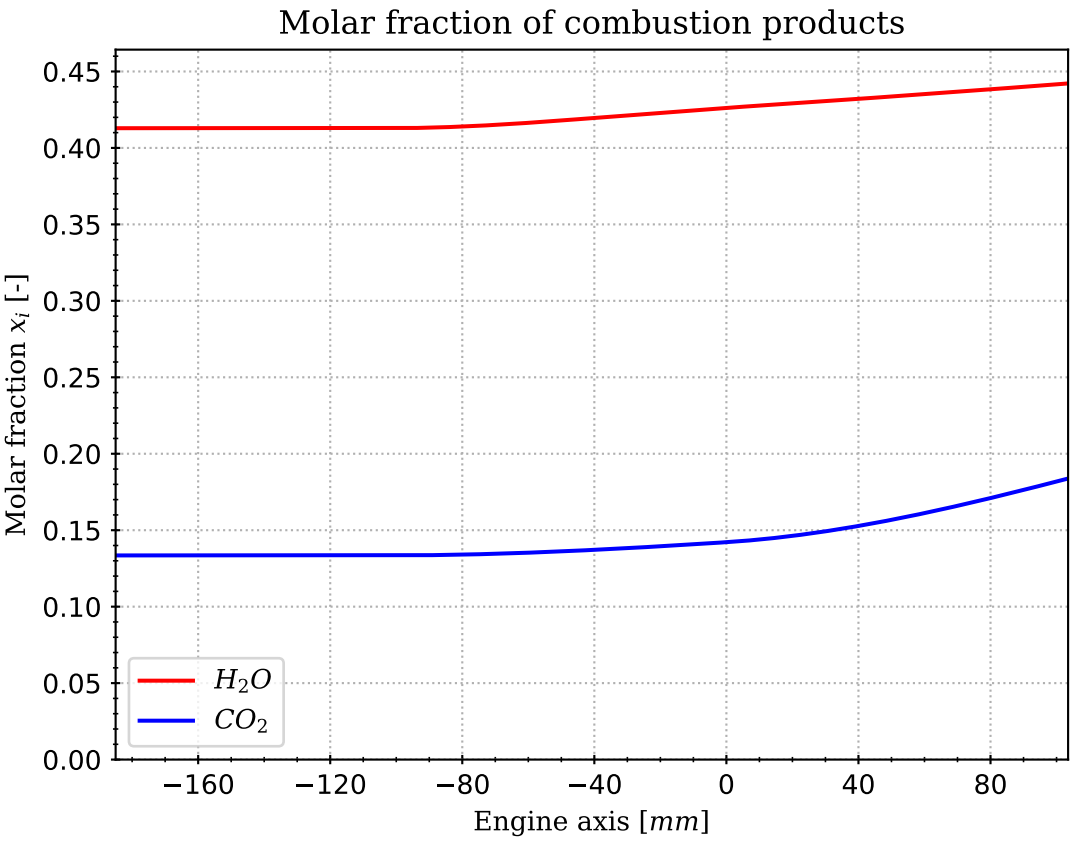

Hot gas conductivity
0.375
0.350
0.325
0.300
0.275
0.250
0.225
0.200
0.175
λ $[\frac{W}{m \cdot K}]$
−160
−120
−80
−40
0
40
80
Engine axis $[mm]$

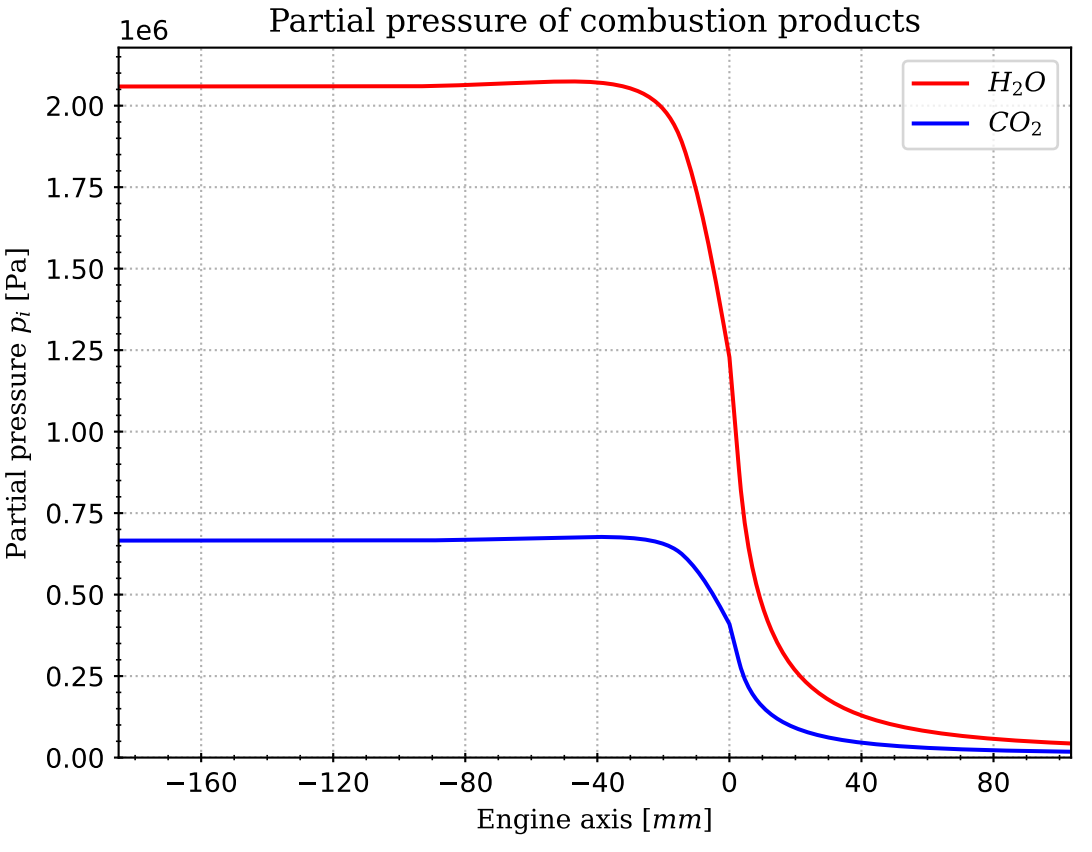

Hot gas Prandtl number
0.720
0.705
0.690
0.675
0.660
0.645
0.630
Pr_g [-]
−160
−120
−80
−40
0
40
80
Engine axis [mm]

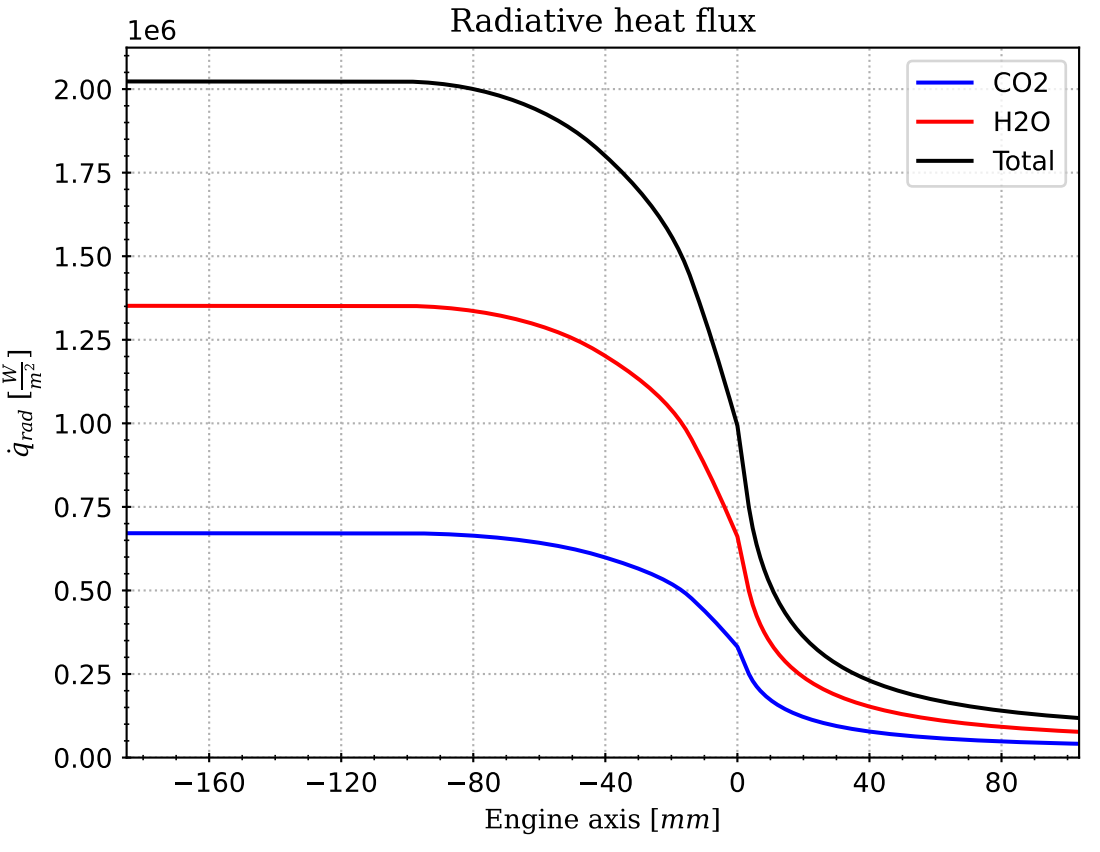

Hot-side convection coefficient h_g
15000
13500
12000
10500
9000
7500
6000
4500
3000
P [Pa]
−160
−120
−80
−40
0
40
80
Engine axis [mm]

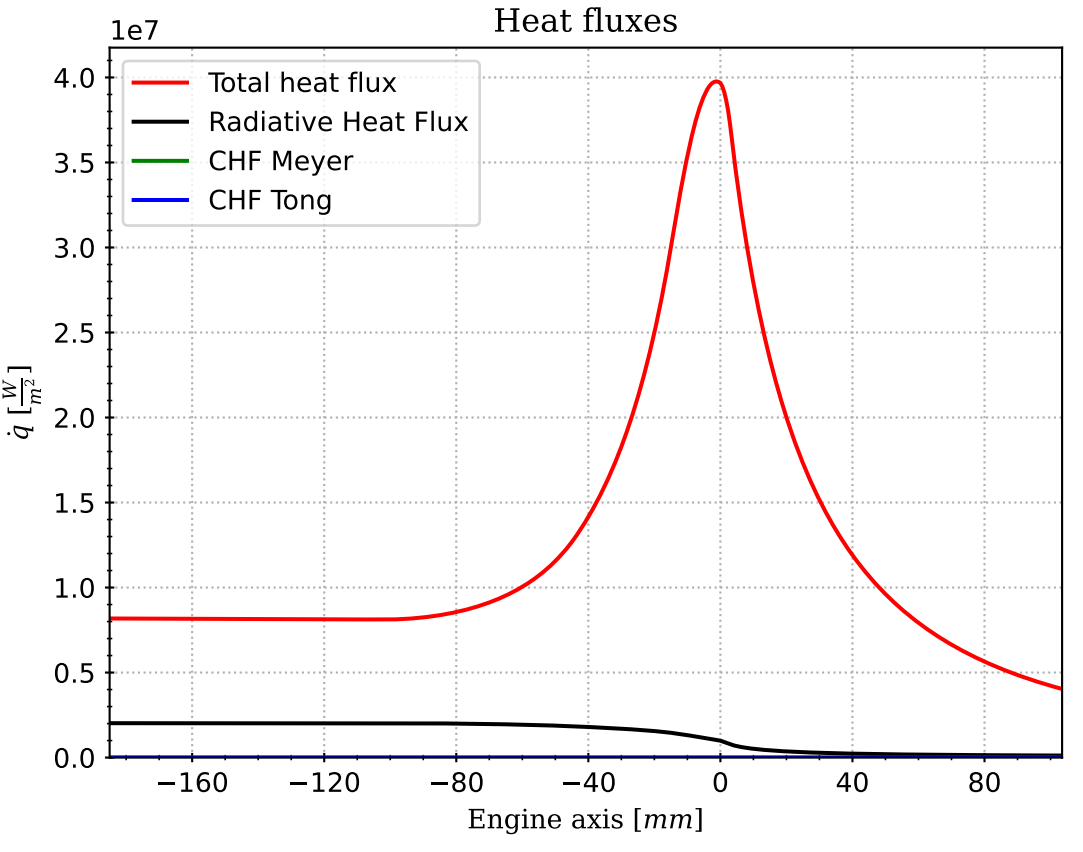

Bartz equation coefficient σ
1.48
1.44
1.40
1.36
1.32
1.28
1.24
1.20
1.16
σ [-]
−160
−120
−80
−40
0
40
80
Engine axis [mm]

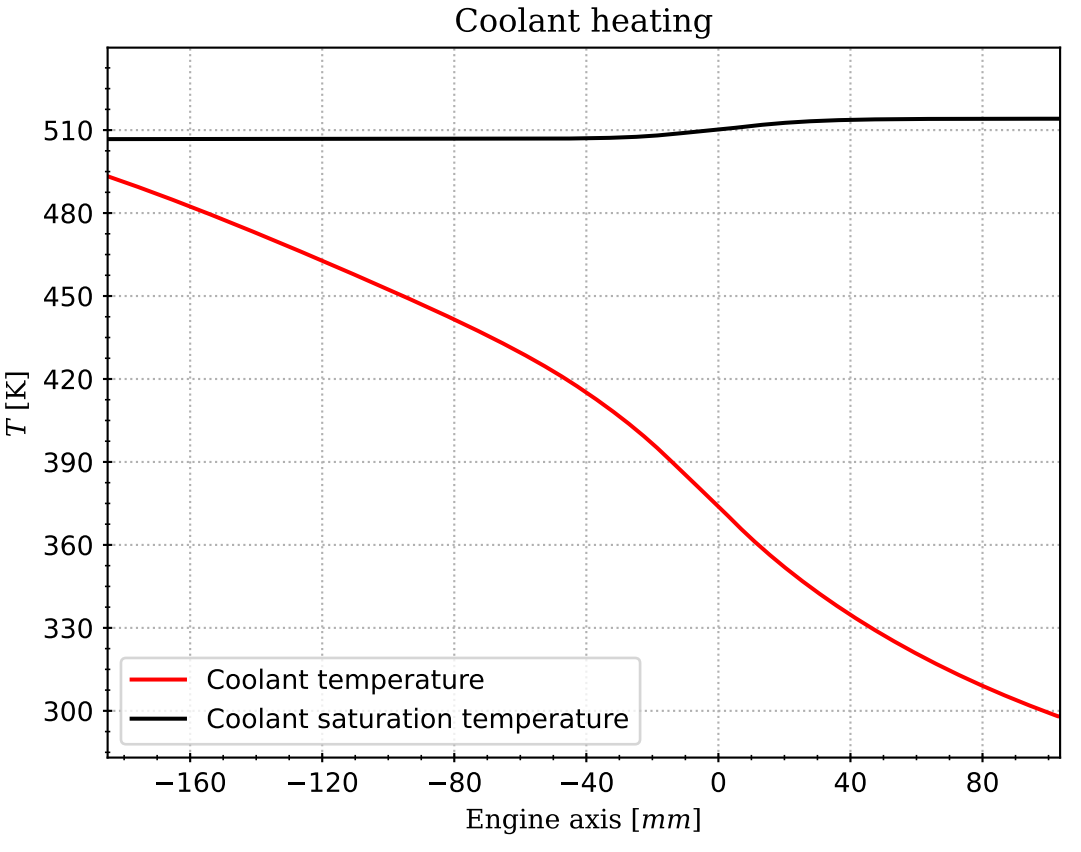

Cold-side convective coefficient h_l/h_{tp}
180000
160000
140000
120000
100000
80000
60000
40000
20000
0
$h\ [\frac{W}{m^2 \cdot K}]$
−160
−120
−80
−40
0
40
80
Engine axis [mm]
Gnielinski
Fin effect correction (Popp & Schmidt)
Subcooled flow boiling (Lui & Winterton)

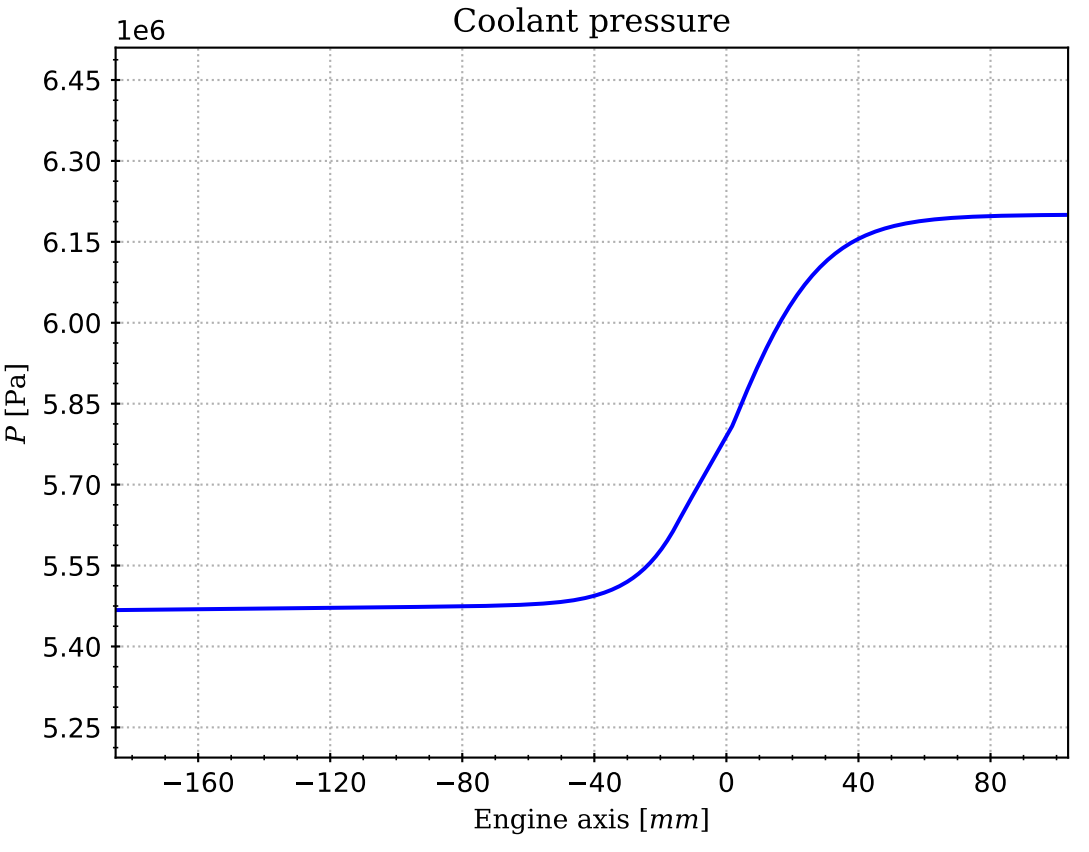

Molar fraction of combustion products
Molar fraction x_i [-]
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−160
−120
−80
−40
0
40
80
Engine axis [mm]
H_2O
CO_2

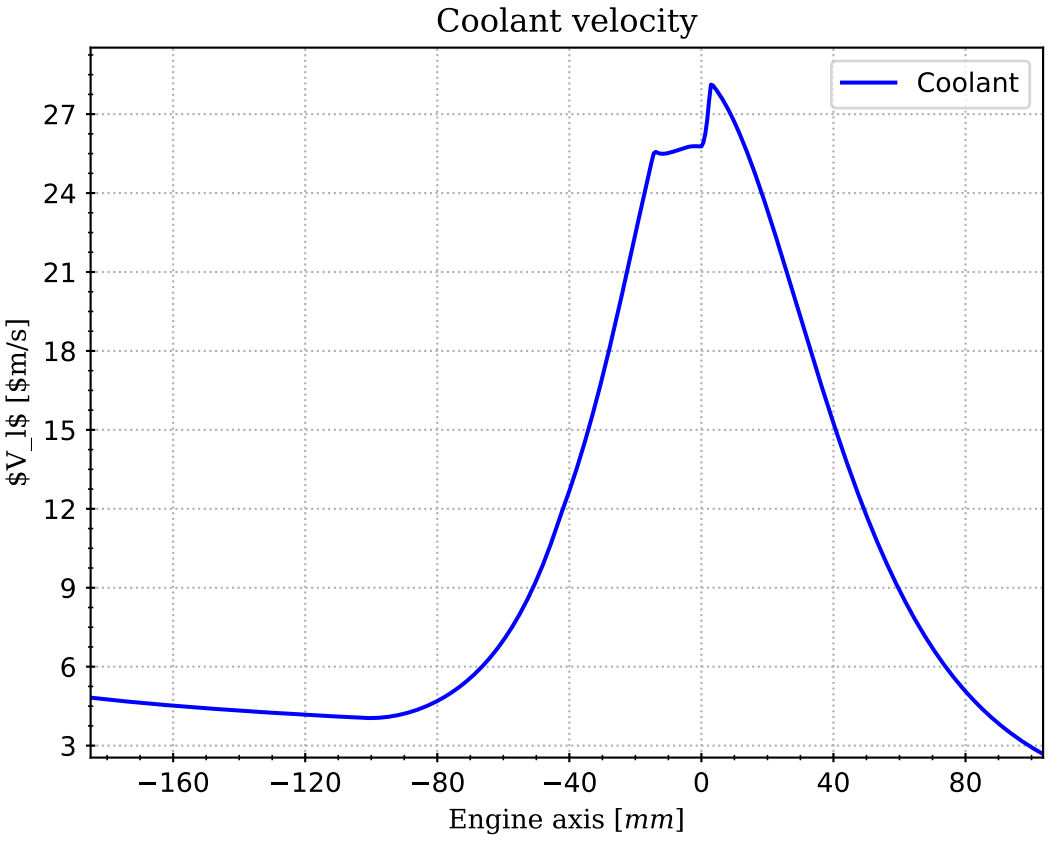

Partial pressure of combustion products
1e6
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Partial pressure p_i [Pa]
−160
−120
−80
−40
0
40
80
Engine axis [mm]
H_2O
CO_2

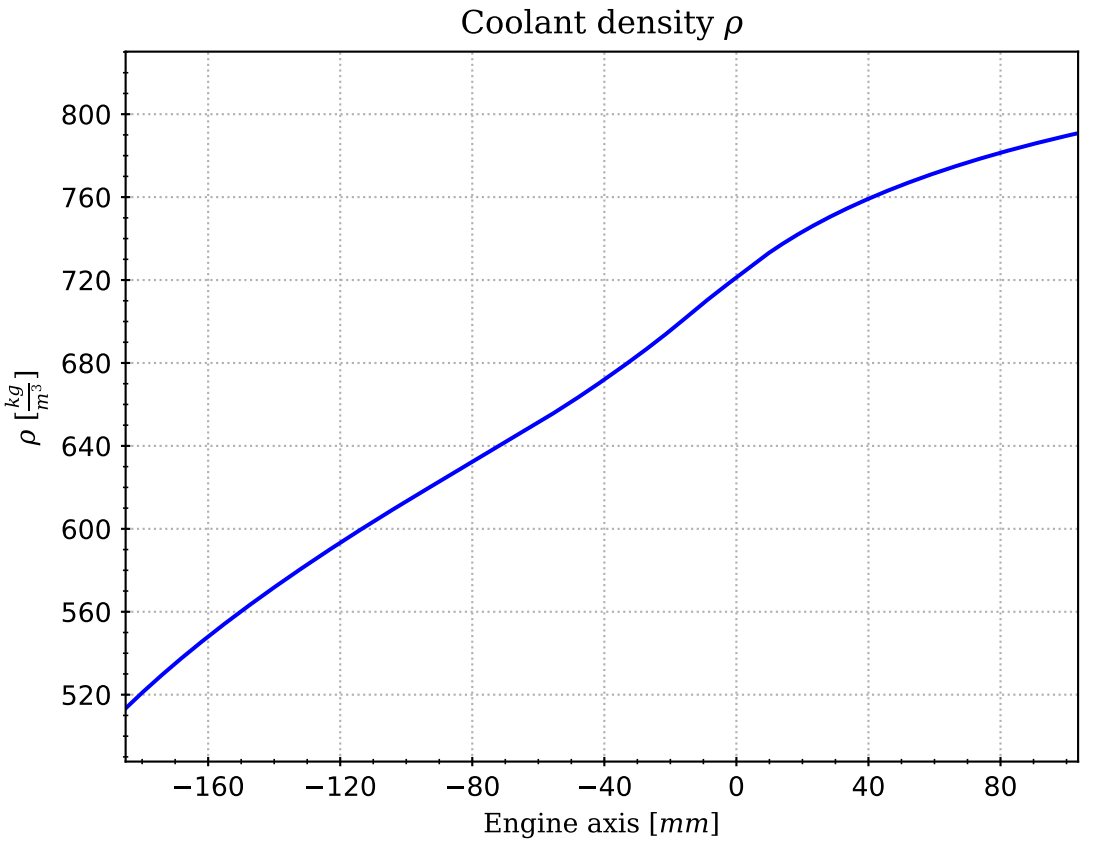

Radiative heat flux
1e6
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
$\dot{q}_{rad}$ $[\frac{W}{m^2}]$
−160
−120
−80
−40
0
40
80
Engine axis $[mm]$
CO2
H2O
Total

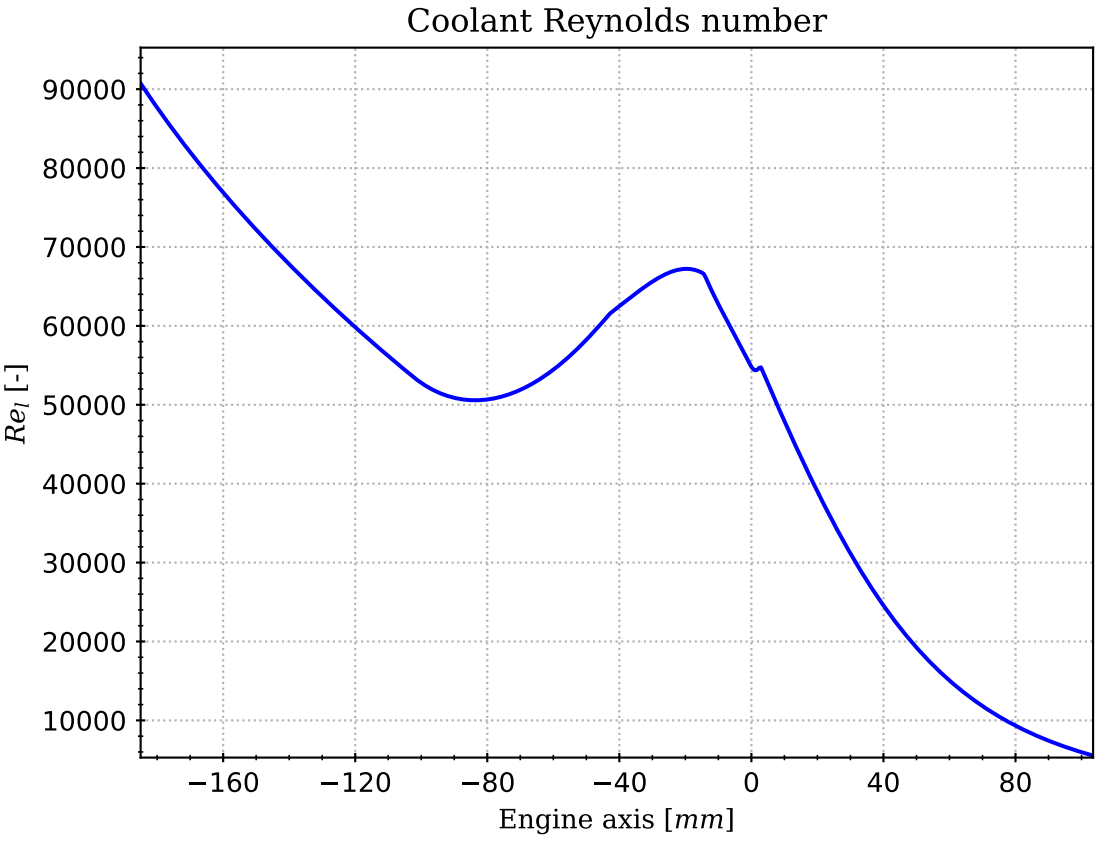

Heat fluxes
1e7
Total heat flux
Radiative Heat Flux
CHF Meyer
CHF Tong
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
$\dot{q}\ [\frac{W}{m^2}]$
−160
−120
−80
−40
0
40
80
Engine axis [mm]

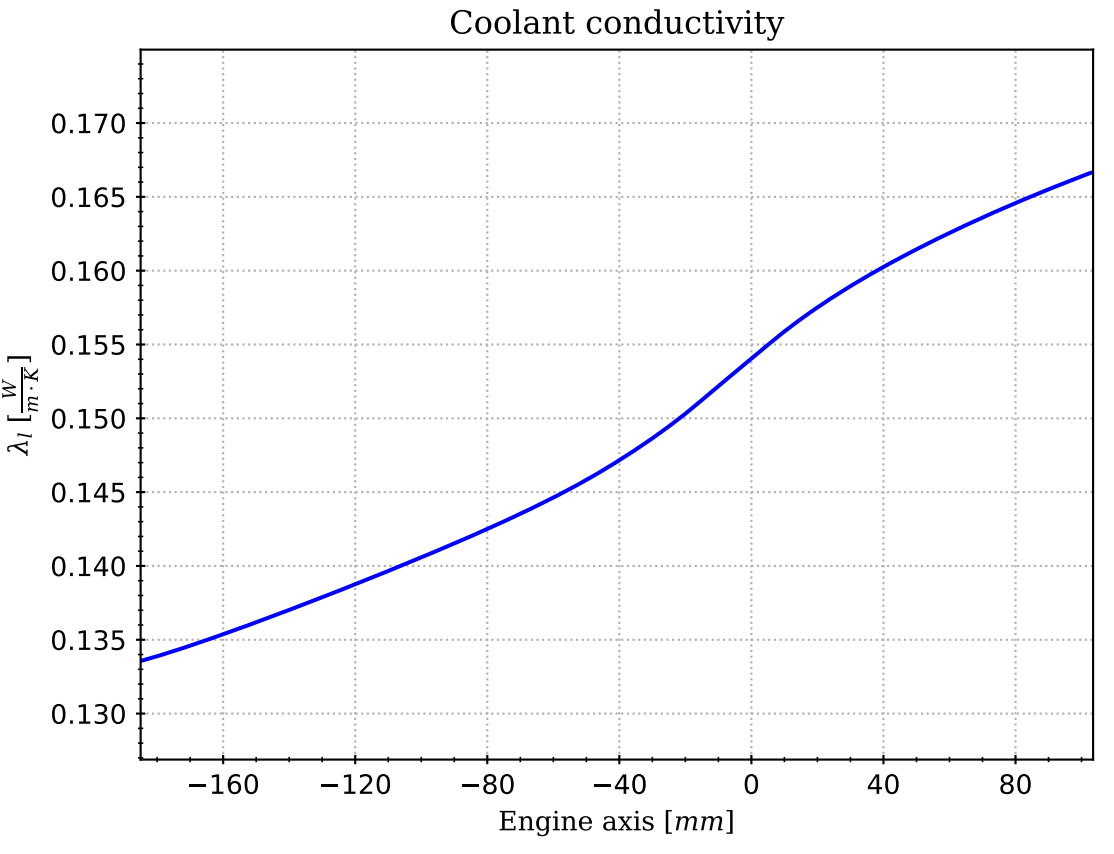

Coolant heating
510
480
450
420
390
360
330
300
T [K]
−160
−120
−80
−40
0
40
80
Engine axis [mm]
Coolant temperature
Coolant saturation temperature

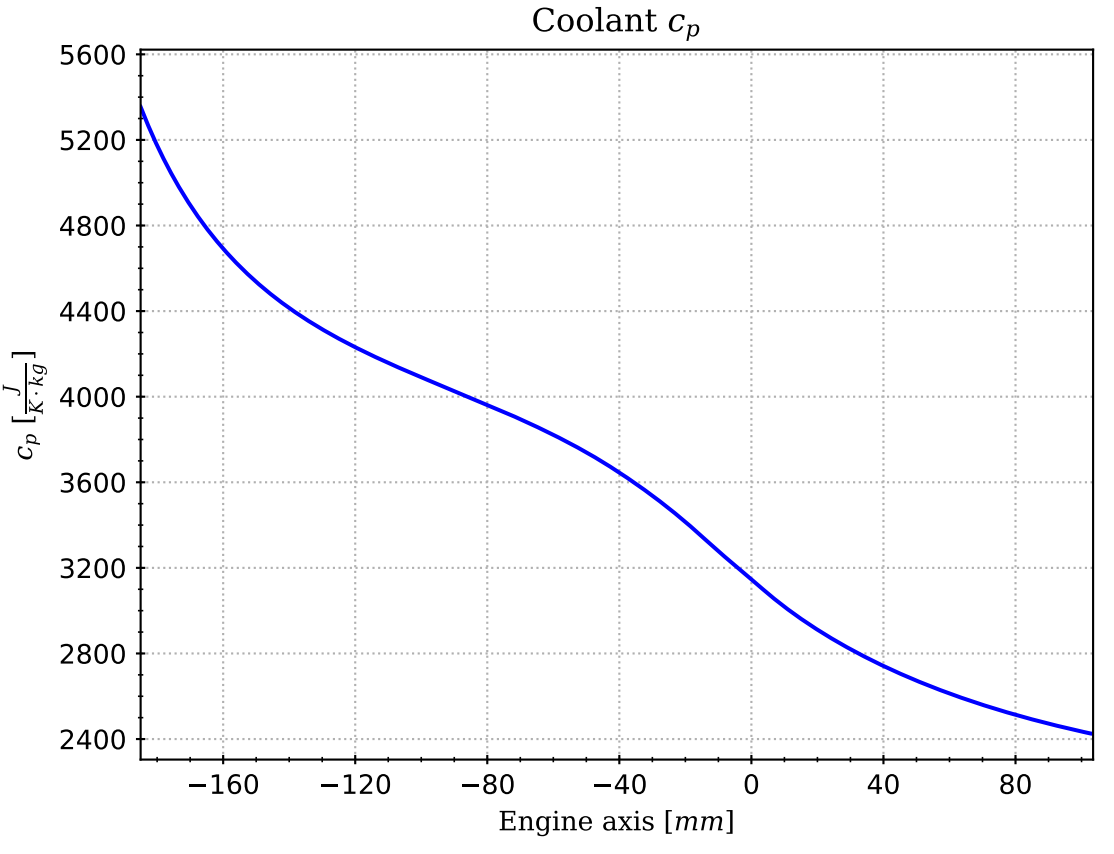

Coolant pressure
1e6
6.45
6.30
6.15
6.00
5.85
5.70
5.55
5.40
5.25
P [Pa]
−160
−120
−80
−40
0
40
80
Engine axis [mm]

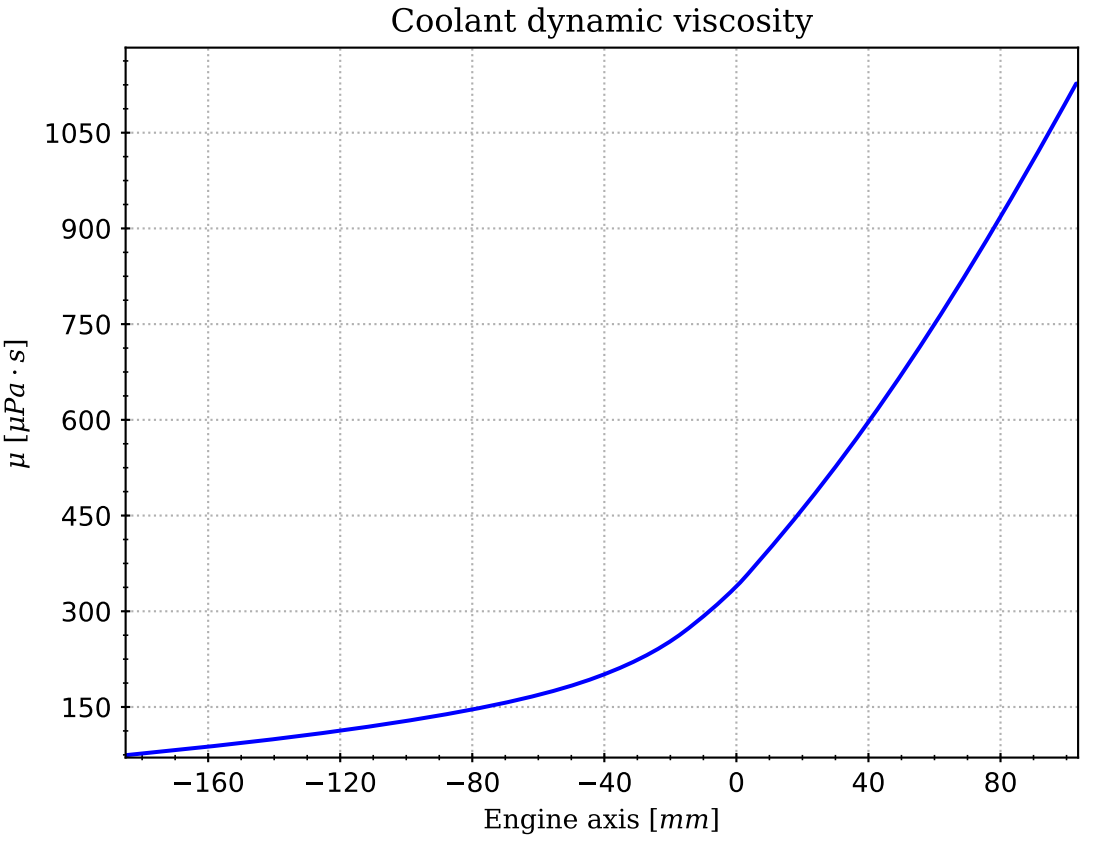

Coolant velocity
Coolant
V_l [$m/s]
27
24
21
18
15
12
9
6
3
−160
−120
−80
−40
0
40
80
Engine axis [mm]

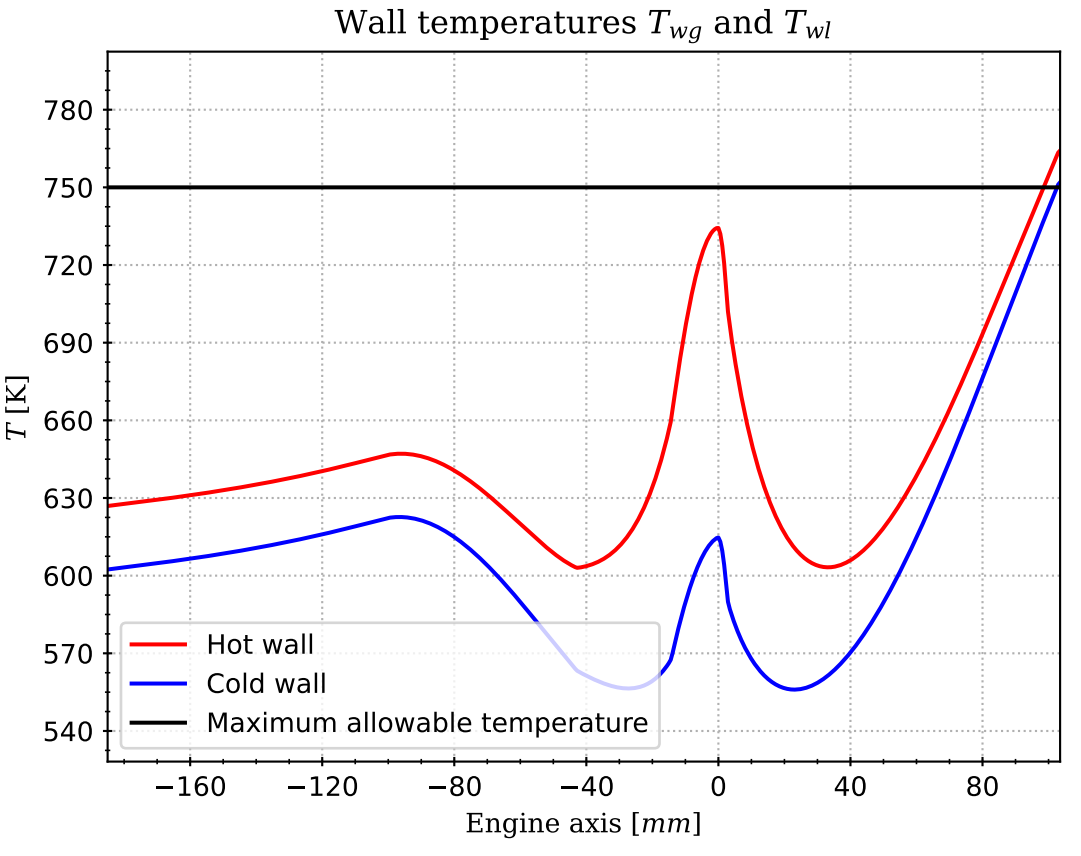

Coolant density ρ
800
760
720
680
640
600
560
520
ρ [$\frac{kg}{m^3}$]
−160
−120
−80
−40
0
40
80
Engine axis [mm]

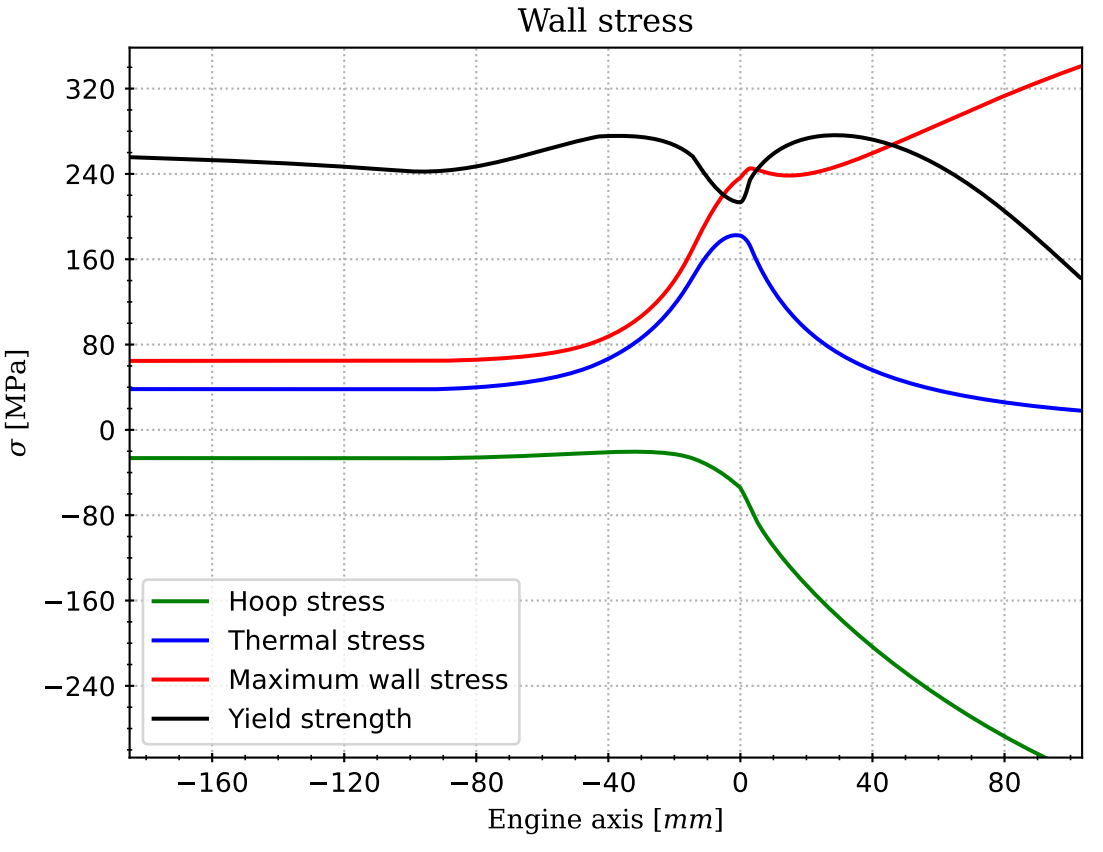

Coolant Reynolds number
90000
80000
70000
60000
50000
40000
30000
20000
10000
Re_l [-]
−160
−120
−80
−40
0
40
80
Engine axis [mm]

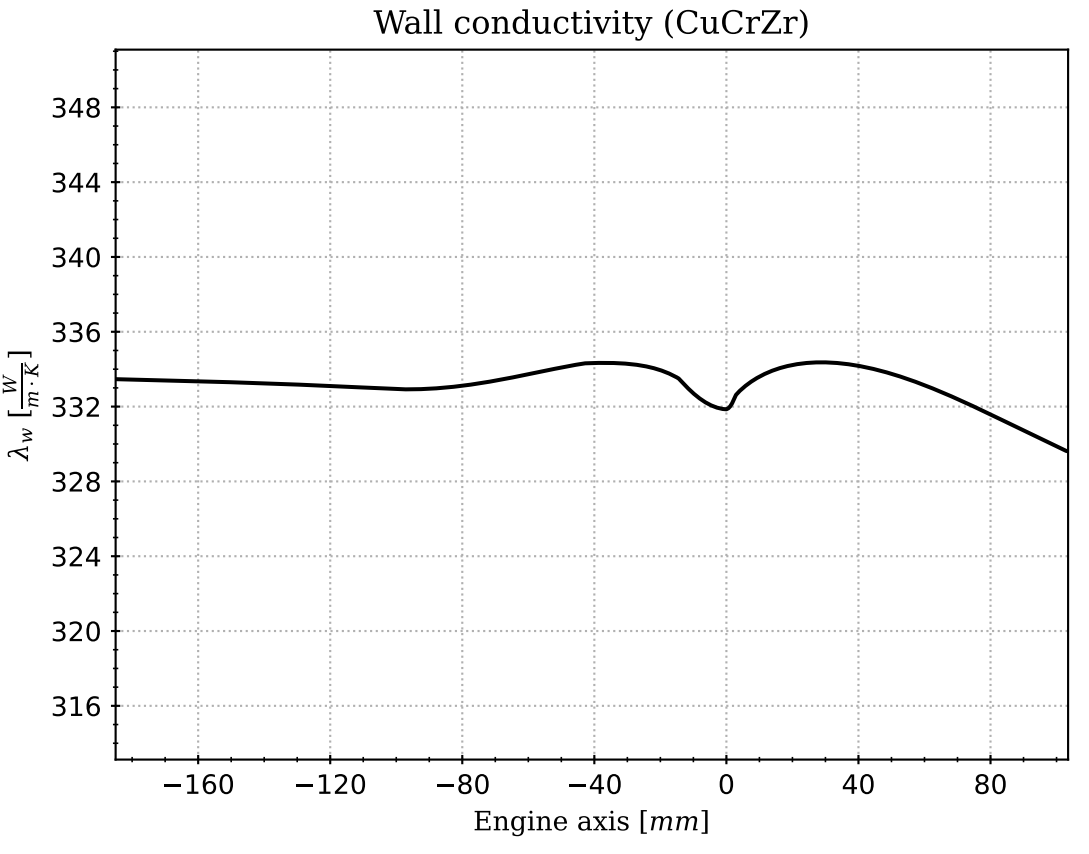

Coolant conductivity
0.170
0.165
0.160
0.155
0.150
0.145
0.140
0.135
0.130
λ_l $[\frac{W}{m \cdot K}]$
−160
−120
−80
−40
0
40
80
Engine axis $[mm]$


Coolant c_p
5600
5200
4800
4400
4000
3600
3200
2800
2400
c_p $[\frac{J}{K \cdot kg}]$
−160
−120
−80
−40
0
40
80
Engine axis $[mm]$


Coolant dynamic viscosity
1050
900
750
600
450
300
150
μ [$\mu Pa \cdot s$]
−160
−120
−80
−40
0
40
80
Engine axis [mm]


Wall temperatures T_{wg} and T_{wl}
780
750
720
690
660
630
600
570
540
T [K]
−160
−120
−80
−40
0
40
80
Engine axis [mm]
Hot wall
Cold wall
Maximum allowable temperature


Wall stress
σ [MPa]
320
240
160
80
0
−80
−160
−240
−160
−120
−80
−40
0
40
80
Engine axis [mm]
Hoop stress
Thermal stress
Maximum wall stress
Yield strength


Wall conductivity (CuCrZr)
348
344
340
336
332
328
324
320
316
λ_w $[\frac{W}{m \cdot K}]$
−160
−120
−80
−40
0
40
80
Engine axis $[mm]$